

Jio Subscriber Churn Intelligence & Revenue Protection

Application Demonstration Walkthrough

What this app does

A 3-page Streamlit executive application built on top of a SQLite database (**jio_retention.db**) and an Optuna-tuned XGBoost champion model. It lets a business user explore the subscriber base, score any individual subscriber's churn risk live, and ask free-text questions that are answered with real-time SQL queries against the database - not canned responses.

Headline results

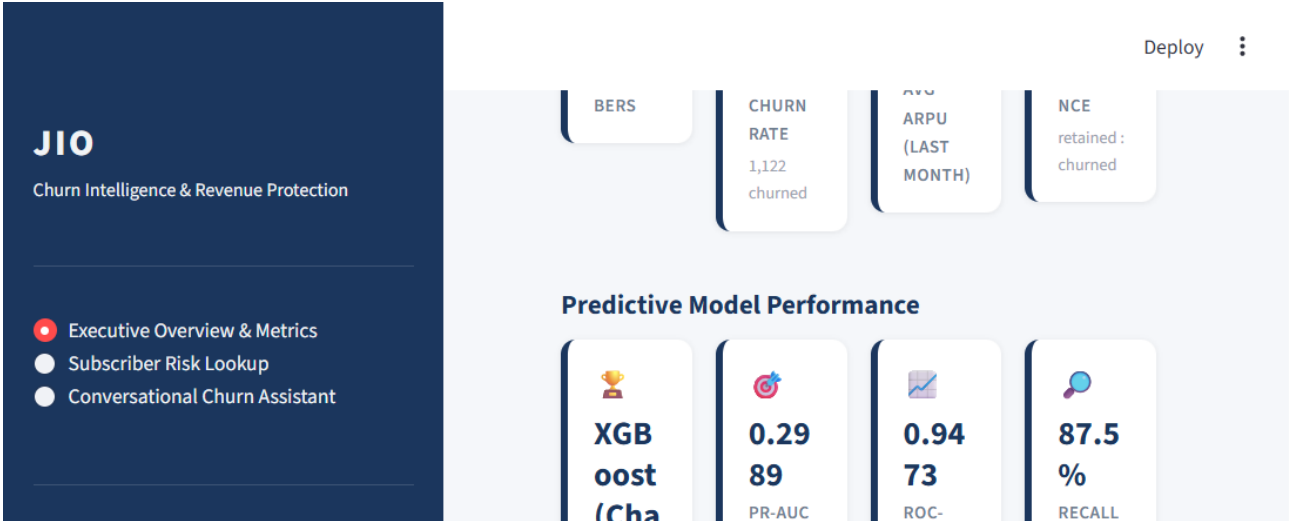
Metric	Value
Subscribers analyzed	64,738
Baseline 30-day churn rate	1.73% (1,122 churned)
Champion model	XGBoost (Optuna-tuned)
ROC-AUC	0.9473
PR-AUC	0.2989
Recall @ Top 10% Risk	87.5%
Top churn signal	MNP (portability) enquiry

The three pages

- 1. Executive Overview & Metrics** - KPI cards, model comparison table, and 6 diagnostic charts.
- 2. Subscriber Risk Lookup** - Search any subscriber_id for a live churn-risk score and plain-English explanation of why they're flagged.
- 3. Conversational Churn Assistant** - Ask natural-language questions; every answer is fetched live via SQL from jio_retention.db, with a strict 'Data not available' guardrail for anything out of scope.

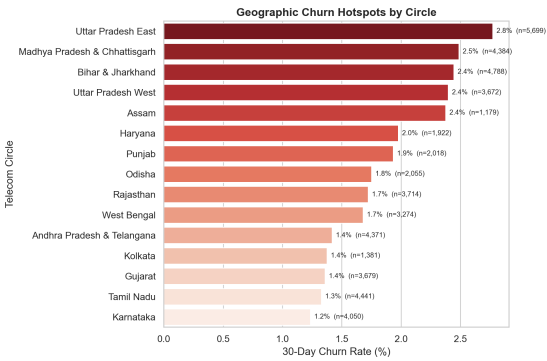
Page 1 - Executive Overview & Metrics

A business snapshot of the subscriber base: dataset KPIs, champion model performance versus the baseline and alternative models, key diagnostic findings, and visual diagnostics - all computed once and cached for instant loading.

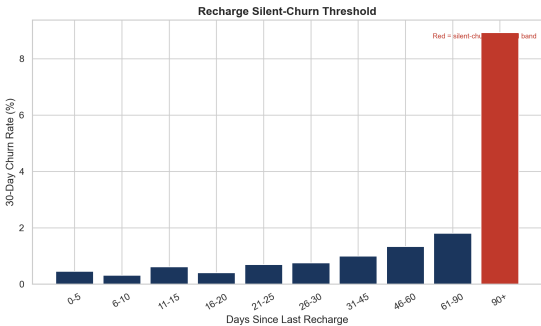


Live screenshot: dataset snapshot KPI cards and predictive model performance cards (champion XGBoost vs. the model comparison table).

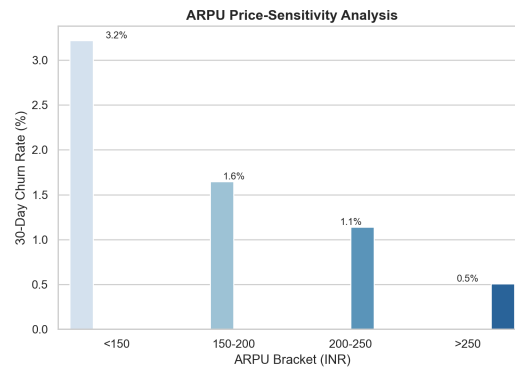
Visual diagnostics generated by the EDA pipeline



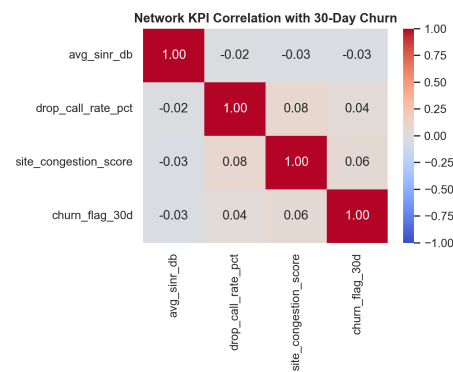
Geographic Churn Hotspots



Recharge Silent-Churn Threshold



ARPU Band Analysis

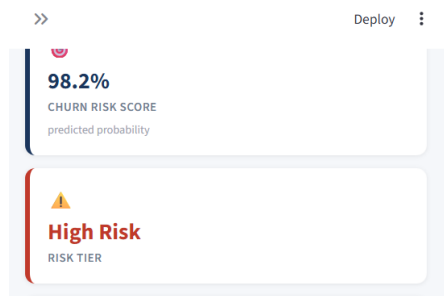


Network KPI Overview

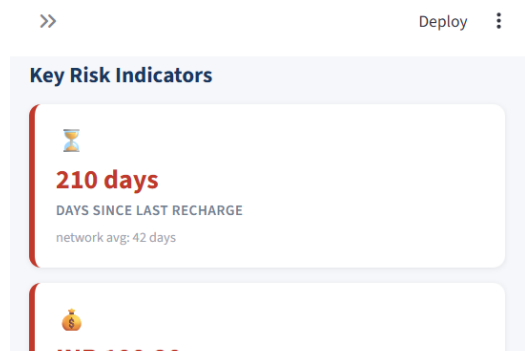
Page 2 - Subscriber Risk Lookup

Enter any **subscriber_id** and get an instant churn-risk score from the persisted XGBoost champion bundle, plus a plain-English rationale mirroring the model's SHAP drivers (MNP enquiry, recharge silence, ARPU decline, competitor usage, unresolved complaints).

Live example: subscriber JIO10022692



Churn Risk Score 98.2%, classified High Risk - and the ground-truth label confirms this subscriber did in fact churn.



Key risk indicator: 210 days since last recharge vs. a network average of 42 days - a textbook 'silent churn' signal.

Why this subscriber was flagged (generated live)

- **MNP enquiry on record** - already checked mobile number portability, the strongest churn signal in the SHAP analysis.
- **Recharge silence** - 210 days since last recharge vs. a 42-day network average.
- **ARPU decline** - last month's spend (INR 100.20) dropped below their own 3-month average (INR 107.45).
- **High competitor usage share** - 57.3% of outgoing usage already routed to competitor networks (network average: 28.2%).

Page 3 - Conversational Churn Assistant

Every answer below was captured live from the running app - no canned responses. Table and column identifiers used in the generated SQL are always validated against the database's own schema (sqlite_master / PRAGMA table_info), never built from raw user text, so the assistant is safe from SQL injection while still answering open-ended questions across every table in the database.

You: how many subscribers in Gujarat

Assistant: There are 3,679 subscribers in Gujarat (5.7% of the 64,738 total).

You: what is the average arpu_last_month_inr

Assistant: The average arpu_last_month_inr in subscribers is 213.70.

You: how many are high risk and from which circle

Assistant: 2,567 subscribers are classified as High Risk (decision threshold 0.84). Note: this dataset's geographic field is circle (telecom circle), not city - breakdown by circle (highest: Uttar Pradesh East, 357 of 5,699 subscribers = 6.3%).

You: what is the capital of France

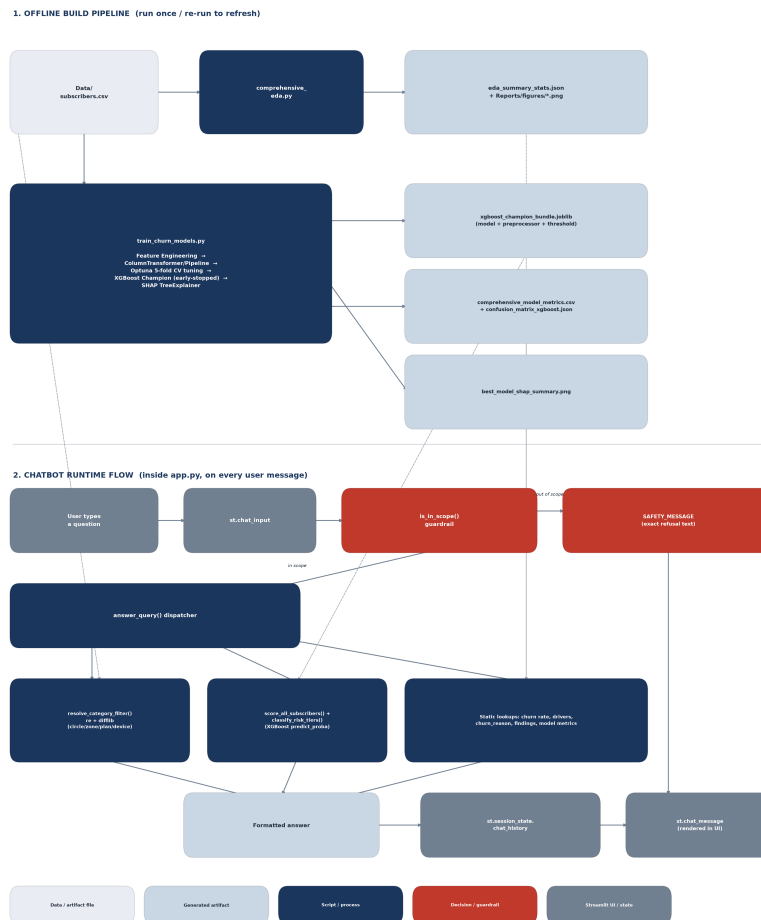
Assistant: I am restricted to answering questions related to Jio subscriber churn intelligence and retention metrics. Data not available for out-of-scope queries.

High Risk subscribers by circle (live SQL + model scoring)

Circle	High Risk Subscribers	Total Subscribers	% of Circle
Uttar Pradesh East	357	5,699	6.3%
Madhya Pradesh & Chhattisgarh	246	4,384	5.6%
Bihar & Jharkhand	245	4,788	5.1%
Uttar Pradesh West	232	3,672	6.3%
Rajasthan	169	3,714	4.6%

System Architecture

Jio Subscriber Churn Intelligence — System & Chatbot Architecture



Tech stack

- **Data layer:** subscribers.csv + Jio_Retention_Dataset.xlsx, compiled into a SQLite database (jio_retention.db) via build_database.py.
- **Modeling:** Logistic Regression baseline, Optuna-tuned XGBoost champion, Random Forest alternative - compared on ROC-AUC, PR-AUC, and Recall@Top-Decile.
- **App:** Streamlit, with st.cache_resource/st.cache_data for the DB connection, schema introspection, and the persisted model bundle.
- **Assistant:** rule-based NL intent resolvers that generate parameterized SQL - counts, aggregates, unique values, top-N breakdowns, and raw-row samples - against any table in the database.

Running it locally

pip install -r requirements.txt → python build_database.py → streamlit run app.py